

Instituto Tecnológico y de
Estudios Superiores de Monterrey.

1. Software Requirements Specification
Software Construction and Decision Making
(Gpo 501)

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THE COWARD KING

Product Backlog / User Stories

Tecnológico de Monterrey

59 User Stories

Section 1 Core Mechanics

User Story #1 Action Points Starting Pool

As	player
I want	start each horde with 5 Action Points (AP)
So that	have resources available to play cards from my very first turn
Acceptance Criteria	
AC1	Player begins every horde with exactly 5 AP.
AC2	AP counter is visible in the HUD at all times.
AC3	AP cannot go below zero.
AC4	Cards requiring more AP than available are visually disabled.
Priority	1 (1 = highest)
Estimation	1 day

User Story #2 AP Carryover Between Turns

As	player
I want	accumulate unused AP across turns with no cap

So that	save resources to deploy powerful multi-card strategies later in the horde
Acceptance Criteria	
AC1	Unspent AP adds to next turn's pool with no upper limit.
AC2	Saved AP persists until the end of the horde.
AC3	HUD displays the updated accumulated AP correctly after each turn.
AC4	AP resets to 5 at the start of the next horde (plus any bonus).
Priority	1 (1 = highest)
Estimation	1 day

User Story #3 AP Bonus on Horde Victory

As	player
I want	receive +5 bonus AP after winning a horde
So that	enter the next horde with a head-start that rewards good play
Acceptance Criteria	
AC1	+5 AP is added to the starting pool of the next horde only.
AC2	Bonus is shown on the horde transition screen.
AC3	HUD reflects the bonus correctly from turn 1 of the next horde.
AC4	Bonus does not stack across multiple horde wins.
Priority	2 (1 = highest)
Estimation	4 hours

User Story #4 King Movement

As	player
I want	move the King one tile in any of 8 directions per turn
So that	reposition him strategically to keep enemies away from his safe zone
Acceptance Criteria	
AC1	Valid adjacent tiles are highlighted when the King is selected.
AC2	King cannot move to tiles occupied by allies or enemies.
AC3	Moving the King consumes the player's main action for that turn.
AC4	The 3x3 safe zone moves together with the King.
Priority	1 (1 = highest)
Estimation	1 day

User Story #5 Dynamic Safe Zone

As	player
I want	have the King's 3x3 safe zone always move with him
So that	maintain consistent protection regardless of where I reposition him
Acceptance Criteria	
AC1	Safe zone is always centered on the King's current tile.
AC2	Safe zone updates visually on every King movement.
AC3	Safe zone is highlighted in a distinct soft-green color on the board.
AC4	Defeat condition checks are always relative to the current safe zone position.
Priority	1 (1 = highest)
Estimation	1 day

User Story #6 Playing Cards from Hand

As	player
I want	spend AP to play one or more cards from my hand in a single turn
So that	deploy multiple units or traps efficiently when I have enough resources
Acceptance Criteria	
AC1	Clicking a card highlights valid placement tiles.
AC2	Multiple cards can be played in one turn if total AP is sufficient.
AC3	Cards cannot be placed on occupied tiles.
AC4	Played cards are removed from hand until the next horde.
Priority	1 (1 = highest)
Estimation	2 days

User Story #7 Turn Phase Structure

As	player
I want	have each turn divided into a clear Player Phase and Enemy Phase
So that	take my action knowing exactly when enemies will move in response
Acceptance Criteria	
AC1	Player Phase always resolves before Enemy Phase.

AC2	During Enemy Phase the player cannot input any action.
AC3	Current phase is indicated in the HUD.
AC4	Turn counter increments only after both phases complete.
Priority	1 (1 = highest)
Estimation	1 day

User Story #8 Enemy Auto-Advance

As	player
I want	see all enemies advance one tile toward the King during every Enemy Phase
So that	predict enemy positions and plan defensive placements accordingly
Acceptance Criteria	
AC1	Every enemy advances exactly 1 tile per turn (Elite Warriors: 2).
AC2	Enemies take the shortest available path toward the King.
AC3	Enemy movement is animated so the player can track it.
AC4	Blocked tiles force enemies to take adjacent detours.
Priority	1 (1 = highest)
Estimation	2 days

User Story #9 Ally Auto-Attack

As	player
I want	have deployed ally units automatically attack enemies within range each Enemy Phase
So that	maintain a defensive line without manually triggering each attack
Acceptance Criteria	
AC1	Each ally attacks the nearest enemy within its defined range every Enemy Phase.
AC2	Damage is applied correctly according to the ally's DMG stat.
AC3	Attack animation plays to give clear visual feedback.
AC4	Allies do not attack if no enemy is in range.
Priority	1 (1 = highest)
Estimation	2 days

User Story #10 End Turn Button

As	player
I want	click an End Turn button to confirm my phase and trigger the Enemy Phase
So that	retain explicit control over when my turn ends, even if I take no action
Acceptance Criteria	
AC1	End Turn button is always visible during the Player Phase.
AC2	Clicking it immediately transitions to Enemy Phase resolution.
AC3	Button is disabled and non-interactive during Enemy Phase.
AC4	Skipping an action and pressing End Turn is always a valid choice.
Priority	1 (1 = highest)
Estimation	4 hours

Section 2 Card System

User Story #11 Random Hand at Horde Start

As	player
I want	receive 3 randomly drawn cards from the pool of 15 at the start of each horde
So that	face a unique strategic challenge every run
Acceptance Criteria	
AC1	Exactly 3 cards are drawn randomly when a horde starts.
AC2	No duplicate cards appear in the same 3-card draw.
AC3	Cards are drawn from the full pool of 15.
AC4	Hand is displayed in the Hand Zone at the bottom of the screen.
Priority	1 (1 = highest)
Estimation	1 day

User Story #12 Card Retention Between Hordes

As	player
I want	choose 1 card from my previous hand to keep when transitioning to the next horde
So that	build continuity between encounters and develop multi-horde strategies
Acceptance Criteria	
AC1	Card retention screen appears after every horde victory.
AC2	All cards from the previous hand are displayed as selectable options.
AC3	Player must choose exactly 1 card to retain.
AC4	Retained card appears in the new hand alongside 3 freshly drawn cards.
Priority	2 (1 = highest)
Estimation	1 day

User Story #13 Maximum Hand Size of 4

As	player
I want	hold a maximum of 4 cards at any time
So that	manage a focused set of tools rather than an overwhelming hand
Acceptance Criteria	
AC1	Hand never exceeds 4 cards under any circumstance.
AC2	Hand Zone renders all 4 cards visibly without overlap.
AC3	Played cards reduce hand size immediately.
AC4	Hovering over a card shows its full stats in a tooltip.
Priority	2 (1 = highest)
Estimation	4 hours

User Story #14 Knight Card [Epic: Attack Units — Story 1 of 4]

As	player
I want	play the Knight card (3 AP) to deploy a melee ally that actively hunts the nearest enemy
So that	have a durable frontline fighter that intercepts threats advancing toward the King

Acceptance Criteria	
AC1	Knight costs exactly 3 AP to place on any unoccupied tile.
AC2	Knight has 80 HP and 30 DMG; stats are displayed on the card face.
AC3	Knight moves 1 tile per turn toward the nearest enemy and attacks when adjacent.
AC4	Knight is removed from the board when its HP reaches 0.
Priority	2 (1 = highest)
Estimation	1 day

User Story #15 Wall Card [Epic: Defense Units — Story 1 of 4]

As	player
I want	play the Wall card (3 AP) to place an immovable high-HP barrier that blocks a tile
So that	force enemies to spend turns destroying or routing around it, buying time for other units
Acceptance Criteria	
AC1	Wall costs exactly 3 AP to place on any unoccupied tile.
AC2	Wall has 150 HP and 0 DMG; stats are displayed on the card face.
AC3	Wall is stationary and does not attack; enemies must either destroy it or path around it.
AC4	Wall is removed from the board when its HP reaches 0.
Priority	2 (1 = highest)
Estimation	4 hours

User Story #16 Exile Card [Epic: Political Cards — Story 1 of 4]

As	player
I want	place the Exile card (2 AP) as a hidden trap tile that paralyzes the first enemy to step on it
So that	neutralize a high-priority threat for 2 turns without spending AP on a combat unit
Acceptance Criteria	
AC1	Exile costs exactly 2 AP to place on any unoccupied tile.

AC2	The first enemy that enters the Exile tile is paralyzed and cannot move for exactly 2 turns.
AC3	Exile is consumed and removed from the board after it triggers once.
AC4	A visual indicator distinguishes the Exile tile from empty tiles while it is active.
Priority	2 (1 = highest)
Estimation	1 day

User Story #17 Royal Guard and Decoy Cards

As	player
I want	use the Royal Guard to absorb one hit for the King and the Decoy to redirect enemy attention
So that	have last-resort protection tools that can save a run when emergencies arise
Acceptance Criteria	
AC1	Royal Guard (2 AP, 90 HP) follows the King automatically and absorbs the first enemy hit aimed at him.
AC2	Decoy (1 AP, 30 HP) causes all enemies to prioritize it over the King while it is alive.
AC3	Both cards are removed from the board when their HP reaches 0.
AC4	Decoy destruction reverts enemy targeting back to the King.
Priority	2 (1 = highest)
Estimation	1 day

User Story #18 Trench Card

As	player
I want	place a Trench card to slow all enemies adjacent to that tile
So that	create chokepoints that buy time for allies to deal damage
Acceptance Criteria	
AC1	Trench costs 1 AP and has 40 HP and 0 DMG.
AC2	Any enemy adjacent to the Trench can move only 1 tile every 2 turns.
AC3	Effect persists as long as the Trench remains on the board.
AC4	Trench is removed when its HP reaches 0.

Priority	3 (1 = highest)
Estimation	4 hours

User Story #19 Card AP Cost Display

As	player
I want	see each card's AP cost displayed prominently on its face in the Hand Zone
So that	make fast resource decisions without needing to open a details panel
Acceptance Criteria	
AC1	AP cost is shown as a number on the top-left corner of each card.
AC2	Cards with insufficient AP are visually grayed out or disabled.
AC3	Cost reflects any permanent upgrades applied to that card.
AC4	Cost is consistent across the Hand Zone, Shop, and hover tooltip.
Priority	2 (1 = highest)
Estimation	4 hours

User Story #20 Card Placement Validation

As	player
I want	receive clear visual feedback distinguishing valid from invalid placement tiles before I click
So that	avoid accidental misplacements and plan my deployment confidently
Acceptance Criteria	
AC1	Valid tiles highlight in a distinct color when a card is selected.
AC2	Occupied or out-of-range tiles show a different color indicating they are blocked.
AC3	Feedback updates in real time as the cursor moves.
AC4	Highlight disappears immediately when selection is cancelled.
Priority	2 (1 = highest)
Estimation	1 day

Section 3

Enemy System

User Story #21 Enemy Spawn System

As	player
I want	see 3 new enemies spawn on the far end of the board each Enemy Phase
So that	face constant pressure that forces me to keep managing the board actively
Acceptance Criteria	
AC1	Exactly 3 enemies spawn per Enemy Phase at rows 0 or 7.
AC2	Spawn columns are randomized each turn.
AC3	No two enemies can spawn on the same tile in the same phase.
AC4	Spawned enemies are immediately registered in the board grid.
Priority	1 (1 = highest)
Estimation	2 days

User Story #22 Skeleton Enemy (Level 1)

As	player
I want	face Skeleton enemies with predictable, low-difficulty stats in Level 1
So that	learn the core loop before facing tougher enemies in later levels
Acceptance Criteria	
AC1	Skeletons have 50 HP and deal 15 DMG.
AC2	Skeletons advance 1 tile per turn toward the King.
AC3	Skeletons are removed from the board when HP reaches 0.
AC4	Skeletons display with their correct sprite distinct from other enemy types.
Priority	2 (1 = highest)
Estimation	1 day

User Story #23 Ogre Enemy (Level 2)

As	player
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I want	fight tougher Ogre enemies in Level 2 that require more coordinated defenses
So that	be forced to use synergy-based card combinations rather than simple placements
Acceptance Criteria	
AC1	Ogres have 80 HP and deal 25 DMG.
AC2	Ogres move 1 tile per turn.
AC3	Ogres require stronger units or multi-card combos to handle efficiently.
AC4	Ogre sprite is visually distinct from Skeletons and Elite Warriors.
Priority	2 (1 = highest)
Estimation	1 day

User Story #24 Elite Warrior Enemy (Level 3)

As	player
I want	face Elite Warriors in Level 3 who move 2 tiles per turn
So that	be forced to rethink defensive formations that worked in earlier levels
Acceptance Criteria	
AC1	Elite Warriors have 100 HP and deal 35 DMG.
AC2	Elite Warriors move 2 tiles per turn, bypassing slow-movement traps.
AC3	Their increased speed makes them the most dangerous regular enemy.
AC4	Elite Warrior sprite is visually distinct from other enemy types.
Priority	2 (1 = highest)
Estimation	1 day

User Story #25 Boss Summoning Mechanic

As	player
I want	fight bosses that summon an ally, remain stationary while it lives, and advance 1 tile when it dies
So that	experience a new layer of challenge that tests how well I can manage the board under pressure
Acceptance Criteria	
AC1	Boss spawns 1 normal enemy adjacent to itself at combat start.
AC2	Boss does not move while the summoned ally is alive.
AC3	Boss advances exactly 1 tile toward the King each time its ally dies.

AC4	Boss immediately summons a new ally after advancing.
Priority	2 (1 = highest)
Estimation	2 days

User Story #26 Boss Stats per Level

As	player
I want	face progressively harder bosses across the three levels
So that	feel a clear difficulty escalation that keeps each level's boss encounter meaningful
Acceptance Criteria	
AC1	Skeleton King (Level 1): 200 HP / 30 DMG, summons Skeletons.
AC2	Ogre Boss (Level 2): 350 HP / 40 DMG, summons Ogres.
AC3	Brave King (Level 3): 500 HP / 50 DMG, summons Elite Warriors.
AC4	Each boss has a visually distinct sprite from regular enemies.
Priority	2 (1 = highest)
Estimation	1 day

Section 4 Win / Lose Conditions

User Story #27 Horde Victory by Survival

As	player
I want	win a horde automatically after surviving 30 turns
So that	have a clear, attainable goal even when I cannot eliminate all enemies
Acceptance Criteria	
AC1	Turn counter tracks progress toward 30 and is always visible.
AC2	Victory triggers at the end of turn 30 if no defeat condition has occurred.
AC3	Card retention screen appears immediately after horde victory.
AC4	10 gold is awarded upon winning a horde.
Priority	1 (1 = highest)
Estimation	4 hours

User Story #28 Defeat by King Contact

As	player
I want	have a clear Game Over triggered the moment an enemy occupies the King's tile
So that	understand exactly what I am protecting and feel the weight of every enemy advance
Acceptance Criteria	
AC1	Game Over is triggered the moment any enemy occupies the King's tile.
AC2	Check occurs after every individual enemy movement step.
AC3	Game state freezes immediately upon King contact.
AC4	Game Over screen shows level, horde reached, and gold lost.
Priority	1 (1 = highest)
Estimation	4 hours

User Story #29 Defeat by Safe Zone Breach

As	player
I want	have a clear Game Over triggered when 2 or more enemies simultaneously occupy the 3x3 safe zone
So that	understand that clustering enemies near the King is just as fatal as direct contact
Acceptance Criteria	
AC1	Game Over triggers if 2 or more enemies simultaneously occupy the 3x3 zone.
AC2	Check occurs after each enemy movement step within the Enemy Phase.
AC3	Safe zone occupancy is tracked in real time.
AC4	Defeat screen is identical to the King-contact screen.
Priority	1 (1 = highest)
Estimation	4 hours

User Story #30 Turn Counter in HUD

As	player
I want	see the current turn out of 30 displayed in the HUD at all times
So that	make time-sensitive decisions knowing exactly how many turns remain

Acceptance Criteria	
AC1	Counter displays as 'Turn X / 30' in the HUD.
AC2	Counter increments at the end of each full turn cycle.
AC3	Counter resets to 1 at the start of each new horde.
AC4	Counter is always visible alongside AP, gold, level, and horde info.
Priority	1 (1 = highest)
Estimation	4 hours

Section 5 Progression & Roguelite

User Story #31 Gold Income per Horde

User Story #31 Gold Income per Horde	
As	player
I want	earn 10 gold every time I win a horde
So that	accumulate currency to invest in upgrades that strengthen future runs
Acceptance Criteria	
AC1	10 gold is awarded immediately after each horde victory.
AC2	Gold is displayed in the HUD at all times during gameplay.
AC3	Gold total is shown on the horde transition screen.
AC4	Gold persists across hordes within a single run.
Priority	2 (1 = highest)
Estimation	4 hours

User Story #32 Gold Loss on Death

As	player
I want	lose all unspent gold when the King dies
So that	feel pressure to spend wisely, knowing that hoarding gold carries real risk
Acceptance Criteria	

AC1	All unspent gold is removed on death.
AC2	Purchased upgrades are always retained regardless of death.
AC3	Gold loss amount is displayed on the Game Over screen.
AC4	Gold counter resets to 0 when the player restarts from the checkpoint.
Priority	2 (1 = highest)
Estimation	4 hours

User Story #33 Permanent Card Upgrades

As	player
I want	spend gold to permanently upgrade cards between hordes
So that	feel long-term improvement that makes each new attempt stronger than the last
Acceptance Criteria	
AC1	Upgrades are purchased via the Upgrade / Shop Screen between hordes.
AC2	Each upgrade costs 20–50 gold depending on its power.
AC3	Upgrades persist after death and across all future runs.
AC4	Upgraded stats are reflected on the card face immediately.
Priority	2 (1 = highest)
Estimation	2 days

User Story #34 Checkpoint System Per Level

As	player
I want	restart from Horde 1 of the current level when the King dies
So that	feel that failure is fair and never lose more progress than one level's worth of work
Acceptance Criteria	
AC1	Death in Level 2 Horde 3 returns to Level 2 Horde 1, not Level 1.
AC2	Checkpoint is set automatically when starting a new level.
AC3	Game Over screen explains the checkpoint behavior clearly.
AC4	Player cannot manually select a starting level; progression is always sequential.
Priority	2 (1 = highest)

Estimation	1 day
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User Story #35 Auto-Save After Level Completion

As	player
I want	have the game auto-save when I complete all 3 hordes and the boss of a level
So that	close the game and resume from my last completed level without losing progress
Acceptance Criteria	
AC1	Auto-save triggers after defeating the boss of each level.
AC2	Save stores: current level index, gold balance, and all purchased upgrades.
AC3	Resume option on the main menu loads the last auto-save correctly.
AC4	No manual save option is required.
Priority	2 (1 = highest)

User Story #36 Level Progression: 3 Hordes + 1 Boss

As	player
I want	progress through exactly 3 regular hordes followed by 1 boss fight within each level
So that	experience a structured difficulty ramp within every level
Acceptance Criteria	
AC1	Each level contains exactly 3 horde encounters and 1 boss fight.
AC2	Boss encounter unlocks only after all 3 hordes of that level are defeated.
AC3	Total encounters across the full game equal 12.
AC4	Level and horde indicator in the HUD reflects the current encounter accurately.
Priority	1 (1 = highest)
Estimation	1 day

Section 6 UI & Screens

User Story #37 Main Menu Screen

As	player
I want	see a main menu when I launch the game with options to start, resume, view stats, and access settings
So that	navigate the game clearly from the first screen without confusion
Acceptance Criteria	
AC1	Main menu shows: Start New Game, Resume, Stats, Options, Credits.
AC2	Resume is grayed out or hidden when no save file exists.
AC3	Menu is visually themed to match the game's medieval aesthetic.
AC4	All buttons navigate to their correct screens.
Priority	2 (1 = highest)
Estimation	2 days

User Story #38 Game Board Screen Layout

As	player
I want	see the 8x8 board centered on screen with the Hand Zone below and HUD above
So that	have all critical information visible without opening any sub-menus
Acceptance Criteria	
AC1	Board Zone occupies the center of the screen.
AC2	Hand Zone shows up to 4 cards at the bottom.
AC3	HUD is shown at the top with turn, AP, gold, level, and horde info.
AC4	All zones are non-overlapping and clearly delineated.
Priority	1 (1 = highest)
Estimation	3 days

User Story #39 Game Over Screen

As	player
I want	see a Game Over screen that shows my progress and lets me retry or quit
So that	quickly understand what I lost and decide my next action without frustration
Acceptance Criteria	

AC1	Screen displays: level reached, horde reached, gold lost, upgrades retained.
AC2	Two options are shown: Retry from Horde 1 of current level, Return to Main Menu.
AC3	Screen is triggered by both defeat conditions.
AC4	Retained upgrades are listed to soften the loss.
Priority	2 (1 = highest)
Estimation	1 day

User Story #40 Horde Transition Screen

As	player
I want	see a transition screen between hordes where I select 1 card to retain
So that	make a meaningful deck-building decision that shapes my next encounter strategy
Acceptance Criteria	
AC1	Screen appears automatically after every horde victory.
AC2	All cards from the previous hand are shown as selectable options.
AC3	Player must confirm selection before proceeding.
AC4	Selected card appears alongside 3 new randomly drawn cards.
Priority	2 (1 = highest)
Estimation	1 day

User Story #41 Upgrade / Shop Screen

As	player
I want	open an upgrade shop between hordes to spend gold on permanent card improvements
So that	invest in my deck during natural breaks without disrupting the game flow
Acceptance Criteria	
AC1	All 15 cards are displayed with current stats and available upgrades.
AC2	Each upgrade shows exact cost and stat change before purchase confirmation.
AC3	Shop is accessible between hordes and during the Player Phase at no turn cost.
AC4	Purchased upgrades are reflected immediately on the card face.
Priority	2 (1 = highest)

Estimation	2 days
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User Story #42 HUD Displays

As	player
I want	see AP, gold, current turn, level, and horde number in the HUD at all times
So that	make informed decisions every turn without opening any extra panel
Acceptance Criteria	
AC1	AP, gold, turn counter, level indicator, and horde indicator are always visible.
AC2	All values update in real time after any relevant game event.
AC3	HUD is legible and does not overlap with the board or hand zone.
AC4	Zero values display as '0', never as empty space.
Priority	1 (1 = highest)
Estimation	1 day

User Story #43 Pause Menu During Gameplay

As	player
I want	pause the game at any time during a horde
So that	take a break or adjust settings without losing my current board state
Acceptance Criteria	
AC1	Pause menu is accessible via a button in the HUD.
AC2	Pause menu offers: Resume, Options, Return to Main Menu.
AC3	All game state is preserved while paused.
AC4	Enemy movement is frozen during pause.
Priority	3 (1 = highest)
Estimation	4 hours

User Story #44 Victory Screen

As	player
I want	see a Victory Screen after completing all 12 encounters

So that	feel proper satisfaction and closure at the end of a successful run
Acceptance Criteria	
AC1	Victory Screen triggers only after defeating the Brave King.
AC2	Screen shows a congratulatory message and run summary.
AC3	Summary includes enemies defeated and levels completed.
AC4	Option to return to main menu is provided.
Priority	3 (1 = highest)
Estimation	1 day

Section 7 Technical & Engineering Backlog

User Story #45 Board Grid Data Model [Engineering Backlog]

As	engineering team
I want	define and implement the Board component as an 8x8 grid[row][col] data structure with a single-occupancy rule per tile
So that	provide a reliable, queryable spatial model that all game systems (placement, pathfinding, safe zone, defeat detection) can depend on without duplication or inconsistency
Acceptance Criteria	
AC1	Board exposes placeObject(obj, row, col), removeObject(row, col), getObject(row, col), and isEmpty(row, col) as its public API; no other system modifies grid state directly.
AC2	placeObject() throws a descriptive error and makes no state change if the target tile is occupied or out of bounds.
AC3	Board state is updated synchronously so that any read immediately after a write reflects the new state.
AC4	All internal coordinates use a consistent (row: 0–7, col: 0–7) convention; no magic index arithmetic appears outside the Board class.
Priority	1 (1 = highest)
Estimation	2 days

User Story #46 Core Class Hierarchy [Engineering Backlog]

User Story #46 Core Class Hierarchy [Engineering Backlog]	
As	engineering team
I want	define and implement the full class hierarchy — GameObject, Unit, Card, Board, TurnManager, HandSystem, and EnemySpawner — with clearly specified interfaces before any feature work begins
So that	ensure every developer on the team works against stable, well-documented contracts so that parallel feature development does not produce conflicting or redundant logic
Acceptance Criteria	
AC1	GameObject is the base class for all entities; it exposes position, type, isActive, update(), render(), and interact() — no derived class bypasses these.
AC2	Unit (extends GameObject) adds HP, DMG, speed, move(), attack(), and takeDamage(amount); all combat entities extend Unit and override only what differs.
AC3	Card (extends GameObject) adds cost, name, cardType, activate(targetTile), and getValidTiles(); all 15 card implementations extend Card with no duplicated placement logic.
AC4	TurnManager, HandSystem, and EnemySpawner are documented with their public method signatures and responsibilities in the codebase before implementation; any cross-system call goes through these interfaces only.
Priority	1 (1 = highest)
Estimation	3 days

User Story #47 Enemy Navigation Around Obstacles

As	player
I want	see enemies intelligently navigate around walls, allies, and other obstacles to always reach the King
So that	face threats that feel relentless and smart, so that placing a Wall or Guardian creates a meaningful tactical decision rather than a permanent blocker
Acceptance Criteria	
AC1	Every enemy resolves its movement by taking the shortest available path to the King's current tile each Enemy Phase; paths update dynamically if the King moves or obstacles change.
AC2	When the direct path is blocked by a Wall, ally, or other enemy, the unit selects the shortest unobstructed adjacent route; it never skips its turn due to a blocked path.
AC3	Elite Warriors apply the same pathfinding logic but advance 2 tiles per resolution step, consuming two path nodes per Enemy Phase.

AC4	Pathfinding produces no out-of-bounds accesses, no infinite loops, and completes within a single frame for all ≤ 24 simultaneously active enemies (3 enemies $\times$ 8 turns max on board).
Priority	2 (1 = highest)
Estimation	3 days

User Story #48 Run Persistence Across Sessions

As	player
I want	be able to close the game at any time and return later to find my checkpoint, gold balance, and purchased upgrades exactly as I left them
So that	play in short sessions without fear of losing hard-earned progress, reinforcing the roguelite loop rather than undermining it
Acceptance Criteria	
AC1	The game writes a save file automatically upon completing each level (after the boss is defeated); no manual save action is required from the player.
AC2	The save file stores: current level index (1–3), current gold balance, and the full list of purchased card upgrades with their stat values.
AC3	On launch, if a valid save file exists, the Resume option on the main menu loads the player directly to Horde 1 of their saved level with all upgrades applied.
AC4	If the save file is missing, empty, or structurally invalid, the game falls back to a default new-game state and logs a descriptive error; it does not crash or display a blank screen.
Priority	2 (1 = highest)
Estimation	2 days

User Story #49 Mouse-Driven Controls

As	player
I want	control the entire game exclusively with the mouse
So that	enjoy an accessible and intuitive control scheme with no keyboard input required
Acceptance Criteria	
AC1	All gameplay actions — card selection, tile placement, King movement, End Turn, and menu navigation — are performed via left-click only.
AC2	Right-click or clicking any non-interactive area cancels the current selection without modifying game state.

AC3	No keyboard input is required at any point during a horde; the game is fully playable with mouse alone.
AC4	The cursor changes to a contextual icon (pointer, crosshair, or cancel) to communicate the current interaction state before the player clicks.
Priority	2 (1 = highest)
Estimation	1 day

User Story #50 Gameplay Performance Targets

As	player
I want	experience smooth, uninterrupted gameplay throughout every horde and Enemy Phase
So that	focus entirely on tactical decisions without stutters or freezes breaking my concentration
Acceptance Criteria	
AC1	The game renders at a stable 60 FPS during all Player Phase interactions (card hover, tile highlight, King movement preview) on a machine with an Intel Core i5 (8th gen or equivalent) CPU and 8 GB RAM.
AC2	Enemy Phase resolution — including movement of up to 24 simultaneous units, pathfinding recalculation, auto-attack damage application, and board re-render — completes in ≤ 500 ms per turn on the same reference hardware.
AC3	The board renders all 64 tiles, up to 24 enemy units, up to 8 ally units, and all active political card zones simultaneously with no dropped frames, as verified by running the worst-case board state in a manual test.
AC4	No memory leak causes the frame rate to degrade by more than 5 FPS over the course of a full 30-turn horde, as measured by profiling tool output.
Priority	2 (1 = highest)
Estimation	Ongoing

Extension of Stories

User Story #51 Archer Card [Epic: Attack Units — Story 2 of 4]

As	player
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I want	play the Archer card (2 AP) to place a stationary ranged unit that covers a 3-tile radius
So that	deal consistent damage to incoming enemies from a safe distance without repositioning
Acceptance Criteria	
AC1	Archer costs exactly 2 AP to place on any unoccupied tile.
AC2	Archer has 50 HP and 20 DMG; stats are displayed on the card face.
AC3	Archer is stationary and auto-attacks the nearest enemy within 3 tiles each Enemy Phase.
AC4	Archer is removed from the board when its HP reaches 0.
Priority	2 (1 = highest)
Estimation	1 day

User Story #52 Mage Card [Epic: Attack Units — Story 3 of 4]

As	player
I want	play the Mage card (4 AP) to deploy a fragile unit that damages all enemies in a cross-AoE pattern
So that	punish enemy clusters with area damage at the cost of committing high AP to a vulnerable unit
Acceptance Criteria	
AC1	Mage costs exactly 4 AP to place on any unoccupied tile.
AC2	Mage has 40 HP and 25 DMG; stats are displayed on the card face.
AC3	Mage attacks all enemies in a cross pattern up to 2 tiles in each cardinal direction each Enemy Phase.
AC4	Mage is removed from the board when its HP reaches 0.
Priority	2 (1 = highest)
Estimation	1 day

User Story #53 Pikeman Card [Epic: Attack Units — Story 4 of 4]

As	player
I want	play the Pikeman card (2 AP) to deploy a fast interceptor that moves 2 tiles per turn
So that	counter fast enemies like Elite Warriors that outrun slower defensive placements
Acceptance Criteria	

AC1	Pikeman costs exactly 2 AP to place on any unoccupied tile.
AC2	Pikeman has 60 HP and 15 DMG; stats are displayed on the card face.
AC3	Pikeman moves 2 tiles per turn toward the nearest enemy and attacks when adjacent.
AC4	Pikeman is removed from the board when its HP reaches 0.
Priority	2 (1 = highest)
Estimation	1 day

User Story #54 Tower Card [Epic: Defense Units — Story 2 of 4]

As	player
I want	play the Tower card (4 AP) to place a stationary high-damage ranged defender with a 4-tile reach
So that	cover a large area of the board and eliminate threats before they approach the safe zone
Acceptance Criteria	
AC1	Tower costs exactly 4 AP to place on any unoccupied tile.
AC2	Tower has 100 HP and 35 DMG; stats are displayed on the card face.
AC3	Tower is stationary and auto-attacks the nearest enemy within 4 tiles each Enemy Phase.
AC4	Tower is removed from the board when its HP reaches 0.
Priority	2 (1 = highest)
Estimation	4 hours

User Story #55 Guardian Card [Epic: Defense Units — Story 3 of 4]

As	player
I want	play the Guardian card (3 AP) to deploy a durable tanky unit that moves to absorb incoming damage
So that	protect the King's safe zone with a mobile front that can reposition as threats shift
Acceptance Criteria	
AC1	Guardian costs exactly 3 AP to place on any unoccupied tile.

AC2	Guardian has 120 HP and 25 DMG; stats are displayed on the card face.
AC3	Guardian moves 1 tile per turn toward the nearest enemy and attacks when adjacent.
AC4	Guardian is removed from the board when its HP reaches 0.
Priority	2 (1 = highest)
Estimation	4 hours

User Story #56 Squire Card [Epic: Defense Units — Story 4 of 4]

As	player
I want	play the Squire card (2 AP) to place a support unit that reduces damage taken by all adjacent allies
So that	extend the lifespan of fragile units like Archers and Mages through passive damage mitigation
Acceptance Criteria	
AC1	Squire costs exactly 2 AP to place on any unoccupied tile.
AC2	Squire has 70 HP and 10 DMG; stats are displayed on the card face.
AC3	Squire reduces all incoming damage to adjacent allies by 50% while it remains on the board.
AC4	Squire is removed from the board when its HP reaches 0.
Priority	2 (1 = highest)
Estimation	1 day

User Story #57 Royal Decree Card [Epic: Political Cards — Story 2 of 4]

As	player
I want	place the Royal Decree card (3 AP) to continuously push all enemies in a 3x3 zone 1 tile back per turn
So that	create persistent breathing room around the King when enemies are clustering near the safe zone
Acceptance Criteria	
AC1	Royal Decree costs exactly 3 AP to place on any unoccupied tile.
AC2	Every enemy within the 3x3 zone centered on the card is pushed 1 tile away from it each Enemy Phase.
AC3	The zone boundary is highlighted on the board while the card is active.

AC4	The push effect applies to all enemies that enter the zone, not only those present at placement.
Priority	2 (1 = highest)
Estimation	1 day

User Story #58 Peace Treaty Card [Epic: Political Cards — Story 3 of 4]

As	player
I want	place the Peace Treaty card (2 AP) to halve the movement speed of all enemies in a 3x3 area
So that	buy extra turns to reposition allies before dangerous enemies reach the safe zone
Acceptance Criteria	
AC1	Peace Treaty costs exactly 2 AP to place on any unoccupied tile.
AC2	Enemies within the 3x3 zone move 1 tile every 2 turns instead of every turn.
AC3	The slowing effect applies to all enemies that enter the zone during the card's active duration.
AC4	The zone boundary is visually distinct from standard tiles while the card is active.
Priority	2 (1 = highest)
Estimation	1 day

User Story #59 Royal Curse Card [Epic: Political Cards — Story 4 of 4]

As	player
I want	place the Royal Curse card (4 AP) to reduce enemy damage output by 50% in a 3x3 area for 5 turns
So that	safely survive waves where multiple enemies are simultaneously threatening the safe zone
Acceptance Criteria	
AC1	Royal Curse costs exactly 4 AP to place on any unoccupied tile.
AC2	All enemies within the 3x3 zone deal 50% less damage for exactly 5 turns after placement.
AC3	The debuff applies to all enemies that enter the zone during its active duration.

AC4	The effect timer counts down correctly and the debuff expires precisely at the end of turn 5.
Priority	2 (1 = highest)
Estimation	1 day

Functional and Non-Functional Requirements Classification

The product backlog was filtered into functional and non-functional requirements according to SCRUM methodology. Functional requirements describe the gameplay, UI, progression, enemy, card, and user-facing systems of the game. Non-functional and technical requirements describe system quality attributes such as architecture, persistence, usability, performance, and maintainability.

Functional Requirements:

User Stories #1–#44, #47, #49, #51–#59

Non-Functional / Technical Requirements:

User Stories #45, #46, #48, #50

Hybrid Requirements:

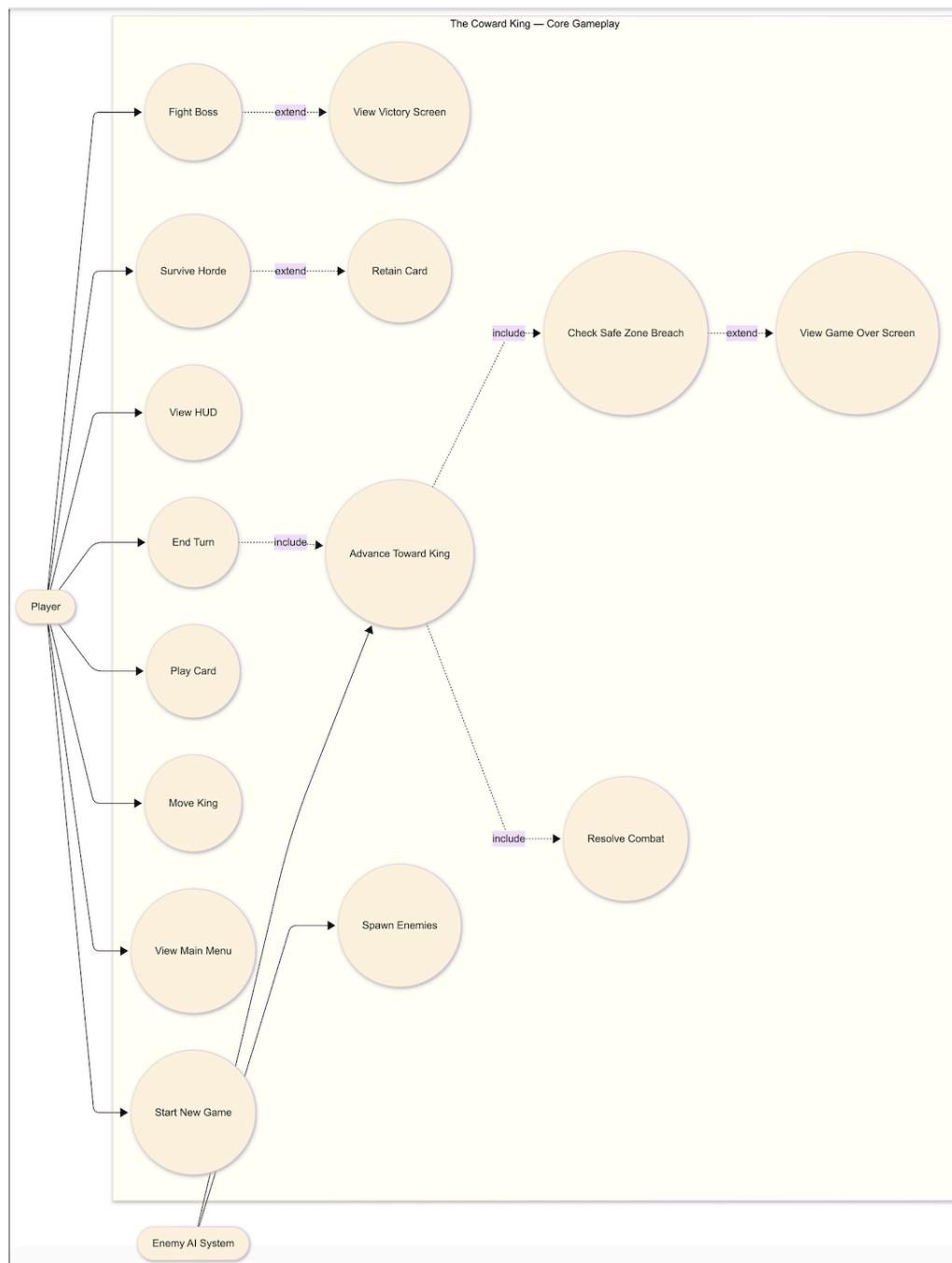
User Story #47 — Enemy Navigation Around Obstacles

User Story #49 — Mouse-Driven Controls

6. UML Use Case Diagrams

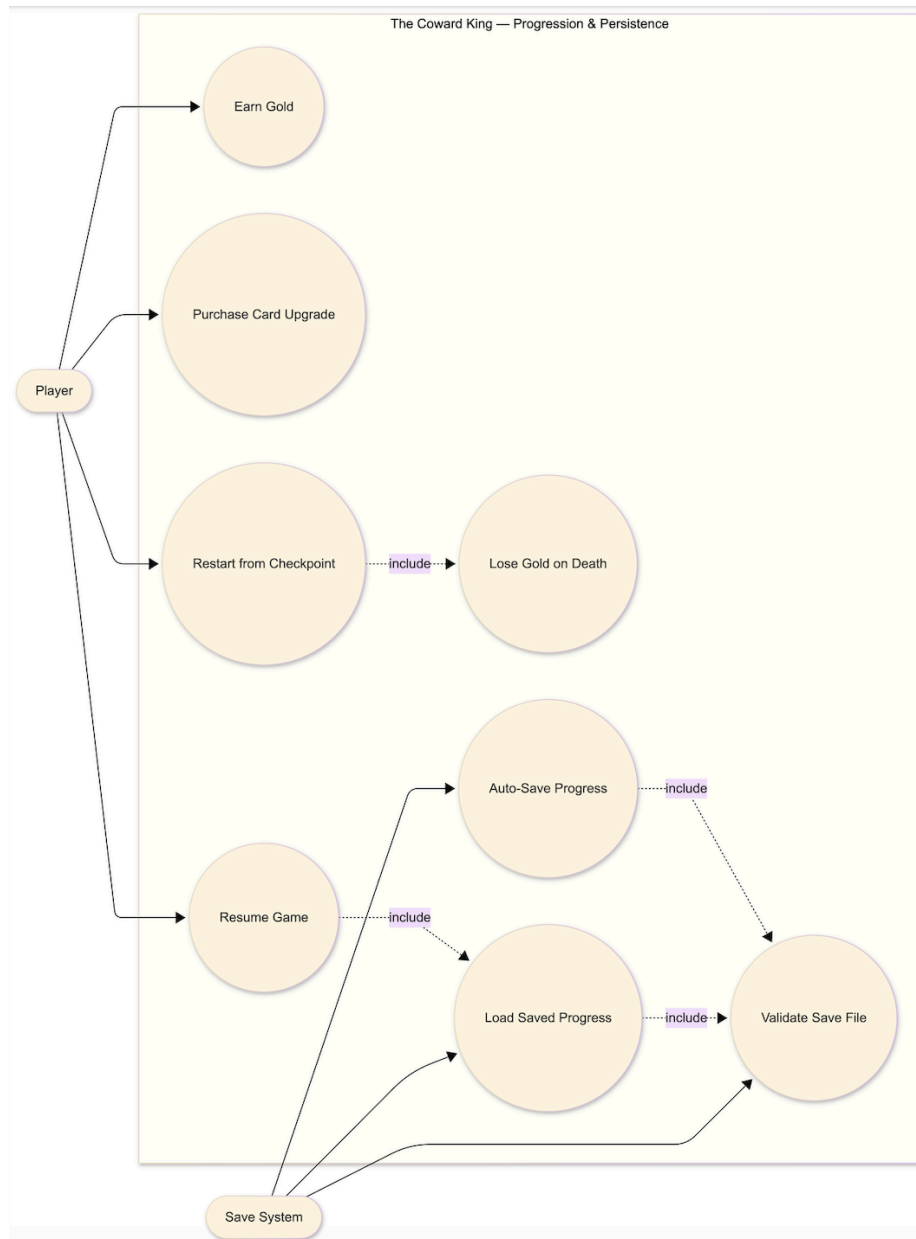
The following UML use case diagrams represent the main interactions between external actors and the proposed system for The Coward King. The diagrams are divided into three areas: core gameplay, progression and persistence, and web/database support. This separation helps keep each diagram clear and focused on a specific part of the system.

6.1 Core Gameplay Use Case Diagram



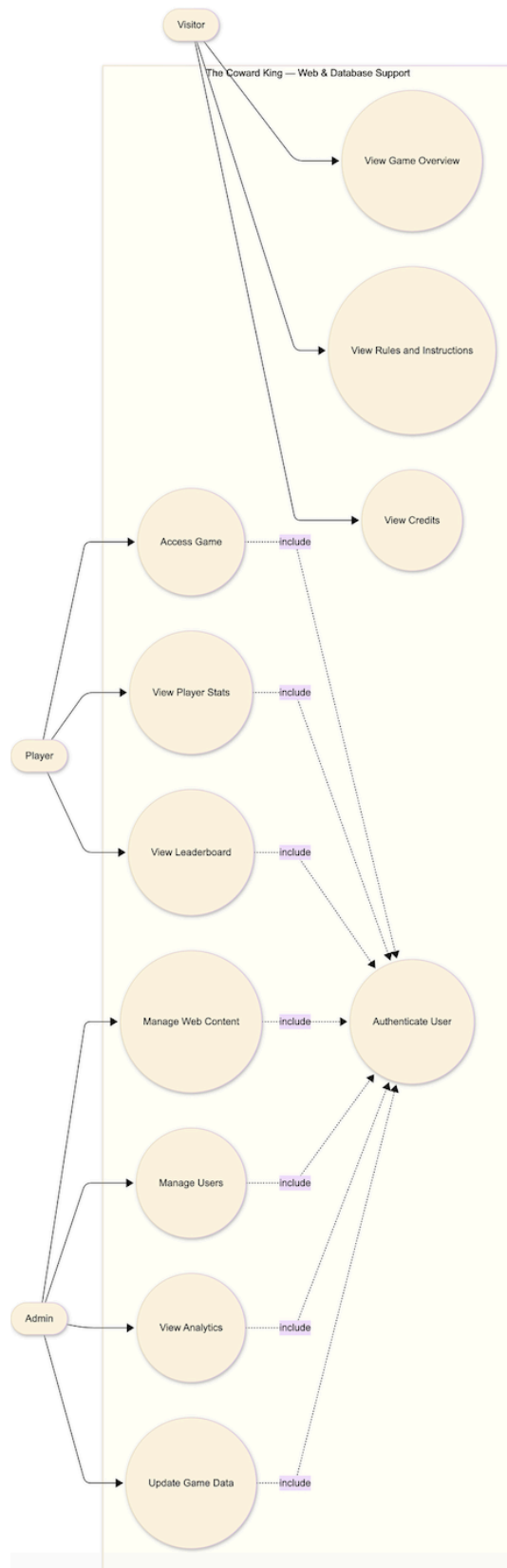
This diagram shows the main gameplay interactions between the player and the core gameplay systems of The Coward King. The player can start the game, view the main menu, move the King, play cards, end the turn, view the HUD, survive hordes, fight bosses, retain cards, and experience game over or victory outcomes. The diagram also includes the Enemy AI System, which is responsible for enemy spawning and enemy advancement toward the King. The include and extend relationships show how some actions depend on or lead to other gameplay events, such as ending the turn triggering enemy advancement, enemy advancement including combat resolution, and safe zone breaches potentially leading to the Game Over screen.

6.2 Progression and Persistence Use Case Diagram



This diagram represents the progression and persistence mechanics of The Coward King. The player can earn gold, purchase card upgrades, resume a saved game, and restart from a checkpoint. The Save System is included as a secondary actor because it handles save-related responsibilities such as auto-saving progress, loading saved progress, and validating save files. This diagram focuses on the systems that allow the player to maintain progress across game sessions and continue developing their run through gold, upgrades, checkpoints, and saved data.

6.3 Web and Database Support Use Case Diagram



This diagram describes the web and database support interactions related to The Coward King. Visitors can view the game overview, rules, instructions, and credits. Players can access the game, view their stats, and view the leaderboard. Admin users can manage web content, manage users,

view analytics, and update game data. The Authenticate User use case is included for actions that require verified access, such as accessing player-specific information or performing admin-level actions. This diagram helps show how the external web platform supports the game through information access, user interaction, and administrative control.

Section 1 — Core Gameplay

Start New Game

UC_Start — Start New Game	
Use Case ID	UC_Start
Use Case Name	Start New Game
Actor(s)	Player
Trigger	Player clicks 'Start New Game' on the Main Menu
Preconditions	Game is launched and Main Menu is displayed
Postconditions	Game initializes Level 1, Horde 1 with 5 AP and a 3-card hand
Main Flow	1. Player clicks Start New Game. 2. System resets all run state (gold, upgrades, level). 3. System draws 3 random cards. 4. System initializes AP to 5. 5. Board is displayed and Player Phase begins.
Alternative Flow	If a save file exists, system prompts player to confirm overwriting current progress.
Exceptions	If initialization fails, system displays an error and returns to Main Menu.

View Main Menu

UC_Menu — View Main Menu	
Use Case ID	UC_Menu
Use Case Name	View Main Menu
Actor(s)	Player
Trigger	Game is launched or player returns from gameplay
Preconditions	Application has loaded successfully
Postconditions	Player selects an option: Start, Resume, Stats, Options, or Credits

Main Flow	1. System displays Main Menu with all options. 2. Resume is grayed out if no save exists. 3. Player selects an option. 4. System navigates to the corresponding screen.
Alternative Flow	If no save file exists, Resume button is disabled.
Exceptions	If the game fails to load assets, an error screen is shown.

Move King

UC_Move — Move King	
Use Case ID	UC_Move
Use Case Name	Move King
Actor(s)	Player
Trigger	Player clicks the King and selects an adjacent tile during Player Phase
Preconditions	It is the Player Phase; King has not yet moved this turn
Postconditions	King moves to selected tile; safe zone updates to new position
Main Flow	1. Player clicks King. 2. System highlights valid adjacent tiles (up to 8 directions). 3. Player clicks a valid tile. 4. King moves to that tile. 5. Safe zone recalculates centered on new position.
Alternative Flow	If player clicks an occupied or invalid tile, move is cancelled and highlights remain.
Exceptions	If King is surrounded on all sides, no valid tiles are shown and move is skipped.

Play Card

UC_PlayCard — Play Card	
Use Case ID	UC_PlayCard
Use Case Name	Play Card
Actor(s)	Player
Trigger	Player clicks a card in the Hand Zone during Player Phase
Preconditions	Player has enough AP; target tile is unoccupied; it is the Player Phase
Postconditions	Card is placed on board; AP is deducted; card is removed from hand
Main Flow	1. Player clicks a card. 2. System highlights valid placement tiles. 3. Player clicks a valid tile. 4. System deducts AP cost. 5. Card unit is placed on board. 6. Card is removed from hand.

Alternative Flow	If AP is insufficient, card is grayed out and cannot be selected.
Exceptions	If no valid tiles exist for placement, system cancels selection and notifies player.

End Turn

UC_EndTurn — End Turn	
Use Case ID	UC_EndTurn
Use Case Name	End Turn
Actor(s)	Player
Trigger	Player clicks the End Turn button
Preconditions	It is the Player Phase
Postconditions	Enemy Phase begins; enemies advance and combat resolves
Main Flow	1. Player clicks End Turn. 2. System disables all player input. 3. Enemy Phase begins: enemies spawn, advance, and combat resolves. 4. Safe zone breach is checked. 5. Turn counter increments. 6. Player Phase resumes.
Alternative Flow	Player may click End Turn without taking any action; this is always valid.
Exceptions	If a defeat condition triggers during Enemy Phase, Game Over screen is shown instead of returning to Player Phase.

View HUD

UC_ViewHUD — View HUD	
Use Case ID	UC_ViewHUD
Use Case Name	View HUD
Actor(s)	Player
Trigger	Any gameplay screen is active
Preconditions	A horde or boss encounter is in progress
Postconditions	Player reads current AP, gold, turn, level, and horde info
Main Flow	1. System renders HUD at top of screen. 2. HUD displays AP, gold, turn counter (X/30), level, and horde number. 3. All values update in real time after any relevant game event.
Alternative Flow	None — HUD is always visible during gameplay.
Exceptions	If a value fails to update, system logs error and forces HUD refresh.

Survive Horde

UC_Survive — Survive Horde	
Use Case ID	UC_Survive
Use Case Name	Survive Horde
Actor(s)	Player
Trigger	Turn counter reaches 30 without a defeat condition occurring
Preconditions	Player is in an active horde encounter
Postconditions	Horde is marked as won; 10 gold awarded; card retention screen appears
Main Flow	1. Turn 30 ends without defeat. 2. System awards 10 gold. 3. System displays horde victory message. 4. Card retention screen is shown (UC_Retain).
Alternative Flow	None.
Exceptions	If gold fails to persist, system retries save and logs error.

Fight Boss

UC_FightBoss — Fight Boss	
Use Case ID	UC_FightBoss
Use Case Name	Fight Boss
Actor(s)	Player
Trigger	Player completes all 3 hordes of a level
Preconditions	All 3 regular hordes of the current level have been won
Postconditions	Boss is defeated or player is defeated; level ends
Main Flow	1. System loads boss encounter for current level. 2. Boss spawns with level-specific stats. 3. Boss summons ally; remains stationary while ally is alive. 4. Player defends for 30 turns or eliminates boss. 5. On boss defeat, Victory Screen is shown.
Alternative Flow	If player dies during boss fight, Game Over screen is shown and checkpoint resets to Horde 1.
Exceptions	If boss AI enters invalid state, system resets boss position and logs error.

Retain Card

UC_Retain — Retain Card

Use Case ID	UC_Retain
Use Case Name	Retain Card
Actor(s)	Player
Trigger	Horde victory triggers card retention screen
Preconditions	Player has won a horde; previous hand cards are available
Postconditions	One card is retained; new hand of 4 cards is formed for next horde
Main Flow	1. System displays all cards from previous hand. 2. Player selects exactly 1 card. 3. Player confirms selection. 4. System draws 3 new random cards. 5. Retained card joins new hand.
Alternative Flow	None — player must select exactly 1 card before proceeding.
Exceptions	If card pool has fewer than 3 remaining cards, system reshuffles available pool.

View Game Over Screen

UC_GameOver — View Game Over Screen	
Use Case ID	UC_GameOver
Use Case Name	View Game Over Screen
Actor(s)	Player
Trigger	A defeat condition is met (King contact or safe zone breach)
Preconditions	An active horde or boss encounter is in progress
Postconditions	Player chooses to retry from checkpoint or return to Main Menu
Main Flow	1. Defeat condition triggers. 2. Game state freezes immediately. 3. System displays Game Over screen with level reached, horde reached, gold lost, and retained upgrades. 4. Player selects Retry or Main Menu.
Alternative Flow	None.
Exceptions	If game state fails to freeze, system forces pause before displaying screen.

View Victory Screen

UC_Victory — View Victory Screen

Use Case ID	UC_Victory
Use Case Name	View Victory Screen
Actor(s)	Player
Trigger	Player defeats the Brave King (final boss of Level 3)
Preconditions	All 12 encounters have been completed successfully
Postconditions	Run summary is displayed; player returns to Main Menu
Main Flow	1. Brave King is defeated. 2. System triggers Victory Screen. 3. Screen shows congratulatory message, enemies defeated, and levels completed. 4. Player clicks Return to Main Menu.
Alternative Flow	None.
Exceptions	None expected; this is a terminal success state.

Spawn Enemies

UC_Spawn — Spawn Enemies	
Use Case ID	UC_Spawn
Use Case Name	Spawn Enemies
Actor(s)	Enemy AI
Trigger	Enemy Phase begins
Preconditions	It is the Enemy Phase of an active horde
Postconditions	3 new enemies are placed on the board at randomized spawn columns
Main Flow	1. Enemy Phase begins. 2. System selects 3 random columns at row 0 or 7. 3. System verifies no two enemies share the same spawn tile. 4. Enemies are placed and registered in the board grid.
Alternative Flow	If a selected column is occupied, system picks an adjacent available column.
Exceptions	If no valid spawn tiles exist, spawn is skipped for that phase and logged.

Advance Toward King

UC_Advance — Advance Toward King

Use Case ID	UC_Advance
Use Case Name	Advance Toward King
Actor(s)	Enemy AI
Trigger	Enemy Phase resolves after spawning
Preconditions	At least one enemy is active on the board
Postconditions	All enemies move 1 tile (Elite Warriors 2 tiles) toward the King via shortest path
Main Flow	1. System iterates over all active enemies. 2. Each enemy calculates shortest path to King. 3. Enemy moves 1 tile (or 2 for Elite Warriors). 4. Combat resolves if enemy is adjacent to an ally. 5. Safe zone breach is checked after each step.
Alternative Flow	If direct path is blocked, enemy takes shortest available detour.
Exceptions	If pathfinding finds no valid route, enemy holds position and logs warning.

Resolve Combat

UC_Combat — Resolve Combat	
Use Case ID	UC_Combat
Use Case Name	Resolve Combat
Actor(s)	Enemy AI
Trigger	An enemy moves adjacent to an ally unit
Preconditions	An enemy and ally occupy adjacent tiles
Postconditions	Damage is applied to both units; units with 0 HP are removed from board
Main Flow	1. Enemy enters range of ally. 2. Ally auto-attacks nearest enemy within range. 3. Damage is applied according to each unit's DMG stat. 4. Attack animations play. 5. Units with 0 HP are removed from board.
Alternative Flow	If no enemy is within ally range, ally does not attack.
Exceptions	If damage calculation produces a negative value, it is clamped to 0.

Check Safe Zone Breach

UC_SafeBreach — Check Safe Zone Breach
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Use Case ID	UC_SafeBreach
Use Case Name	Check Safe Zone Breach
Actor(s)	Game System (internal)
Trigger	After each enemy movement step during Enemy Phase
Preconditions	Enemy Phase is resolving; safe zone coordinates are current
Postconditions	If 2+ enemies occupy the safe zone simultaneously, Game Over is triggered
Main Flow	1. System counts enemies currently inside the 3x3 safe zone. 2. If count ≥ 2 , defeat condition is triggered. 3. Game Over screen is displayed.
Alternative Flow	If count is 0 or 1, no action is taken and Enemy Phase continues.
Exceptions	If safe zone coordinates are stale, system recalculates before checking.

Section 2 — Progression & Persistence

Earn Gold

UC_EarnGold — Earn Gold	
Use Case ID	UC_EarnGold
Use Case Name	Earn Gold
Actor(s)	Player
Trigger	Player wins a horde or boss encounter
Preconditions	Player has successfully completed a horde or boss fight
Postconditions	10 gold is added to the player's current gold balance
Main Flow	1. Horde or boss victory is confirmed. 2. System adds 10 gold to player's balance. 3. Gold total is shown on horde transition screen. 4. HUD updates to reflect new gold amount.
Alternative Flow	None.
Exceptions	If gold fails to persist after victory, system retries and logs error.

Lose Gold on Death

UC_LoseGold — Lose Gold on Death	
Use Case ID	UC_LoseGold
Use Case Name	Lose Gold on Death
Actor(s)	Player
Trigger	King dies triggering a Game Over
Preconditions	Player has unspent gold; a defeat condition has been met
Postconditions	All unspent gold is removed; purchased upgrades are retained
Main Flow	1. Defeat condition triggers. 2. System records gold lost for Game Over screen. 3. System sets gold balance to 0. 4. Purchased upgrades remain unaffected. 5. Game Over screen displays gold lost.
Alternative Flow	None.
Exceptions	If gold reset fails, system forces gold to 0 on next session load.

Purchase Card Upgrade

UC_PurchaseUpgrade — Purchase Card Upgrade	
Use Case ID	UC_PurchaseUpgrade
Use Case Name	Purchase Card Upgrade
Actor(s)	Player
Trigger	Player selects and confirms an upgrade in the Upgrade Shop
Preconditions	Player has sufficient gold; upgrade has not been purchased yet
Postconditions	Upgrade is applied permanently; gold is deducted; card stats update immediately
Main Flow	1. Player opens Upgrade Shop between hordes. 2. System displays all 15 cards with available upgrades and costs. 3. Player selects an upgrade and confirms purchase. 4. System deducts gold cost. 5. Upgrade is applied permanently to the card. 6. Card face updates to reflect new stats.
Alternative Flow	If player lacks sufficient gold, upgrade is grayed out and unselectable.
Exceptions	If upgrade fails to save, system rolls back gold deduction and notifies player.

Restart from Checkpoint

UC_Checkpoint — Restart from Checkpoint	
Use Case ID	UC_Checkpoint
Use Case Name	Restart from Checkpoint
Actor(s)	Player
Trigger	Player selects Retry on the Game Over screen
Preconditions	A defeat condition has been met; checkpoint is set to Horde 1 of current level
Postconditions	Game restarts at Horde 1 of the level where the player died; upgrades are retained; gold resets to 0
Main Flow	1. Player clicks Retry on Game Over screen. 2. System loads checkpoint: Horde 1 of current level. 3. Gold balance resets to 0. 4. All purchased upgrades are applied. 5. New 3-card hand is drawn. 6. Board initializes and Player Phase begins.
Alternative Flow	If player selects Main Menu instead, no checkpoint is loaded.
Exceptions	If checkpoint data is corrupt, system falls back to Level 1 Horde 1 and logs error.

Auto-Save Progress

UC_AutoSave — Auto-Save Progress	
Use Case ID	UC_AutoSave
Use Case Name	Auto-Save Progress
Actor(s)	Save System
Trigger	Player defeats the boss of a level
Preconditions	Boss of current level has been defeated
Postconditions	Save file is written with current level index, gold balance, and all purchased upgrades
Main Flow	1. Boss defeat is confirmed. 2. System collects current level index, gold balance, and upgrade list. 3. System writes save file. 4. Save System validates the written file. 5. Resume option on Main Menu becomes available.
Alternative Flow	None — auto-save triggers automatically with no player action required.
Exceptions	If save write fails, system retries once and logs error; player is notified.

Load Saved Progress

UC_LoadSave — Load Saved Progress	
Use Case ID	UC_LoadSave
Use Case Name	Load Saved Progress
Actor(s)	Save System
Trigger	Player selects Resume on the Main Menu
Preconditions	A valid save file exists on disk
Postconditions	Game loads at Horde 1 of the saved level with all upgrades applied
Main Flow	1. Player clicks Resume. 2. System locates save file. 3. Save file is validated (UC_ValidateSave). 4. System reads level index, gold balance, and upgrade list. 5. Game initializes with loaded state. 6. Board displays and Player Phase begins.
Alternative Flow	None.
Exceptions	If save file is missing or invalid, system falls back to new-game state and logs error.

Resume Game

UC_Resume — Resume Game	
Use Case ID	UC_Resume
Use Case Name	Resume Game
Actor(s)	Player
Trigger	Player clicks Resume on the Main Menu
Preconditions	A valid save file exists; Resume option is not grayed out
Postconditions	Player continues from Horde 1 of their last completed level with all upgrades intact
Main Flow	1. Player clicks Resume. 2. System triggers UC_LoadSave. 3. Save file is loaded and validated. 4. Game state is restored. 5. Player Phase begins at saved level.
Alternative Flow	If no save file exists, Resume is grayed out and cannot be clicked.
Exceptions	If load fails, system returns to Main Menu and displays an error message.

Validate Save File

UC_ValidateSave — Validate Save File	
Use Case ID	UC_ValidateSave
Use Case Name	Validate Save File
Actor(s)	Save System
Trigger	Called during UC_LoadSave or UC_AutoSave
Preconditions	A save file exists or has just been written
Postconditions	Save file is confirmed valid; or fallback to new-game state is triggered
Main Flow	1. System reads save file structure. 2. System checks for required fields: level index, gold balance, upgrade list. 3. System verifies field types and value ranges. 4. If valid, returns parsed save data. 5. If invalid, logs error and triggers fallback.
Alternative Flow	None.
Exceptions	If file is structurally malformed, system discards it and starts new game without crashing.

Section 3 — Web & Database Support

View Game Overview

UC_ViewOverview — View Game Overview	
Use Case ID	UC_ViewOverview
Use Case Name	View Game Overview
Actor(s)	Visitor
Trigger	Visitor navigates to the main web page
Preconditions	Web platform is online and accessible
Postconditions	Visitor reads a summary of The Coward King concept and features
Main Flow	1. Visitor opens the web URL. 2. System displays the game overview page. 3. Visitor reads description, screenshots, and key features.
Alternative Flow	None.
Exceptions	If page fails to load, system shows a generic error page.

View Rules and Instructions

UC_ViewRules — View Rules and Instructions	
Use Case ID	UC_ViewRules
Use Case Name	View Rules and Instructions
Actor(s)	Visitor
Trigger	Visitor clicks on Rules or How to Play section
Preconditions	Web platform is online
Postconditions	Visitor reads full game rules and instructions
Main Flow	1. Visitor clicks Rules link. 2. System displays rules page with AP system, turn structure, card descriptions, win/lose conditions. 3. Visitor reads content.
Alternative Flow	None.
Exceptions	If content fails to load, system displays cached version or error.

View Credits

UC_ViewCredits — View Credits	
Use Case ID	UC_ViewCredits
Use Case Name	View Credits
Actor(s)	Visitor
Trigger	Visitor clicks on Credits section
Preconditions	Web platform is online
Postconditions	Visitor reads team member names and project attributions
Main Flow	1. Visitor clicks Credits link. 2. System displays credits page. 3. Visitor reads team and institutional information.
Alternative Flow	None.
Exceptions	None expected.

Access Game

UC_AccessGame — Access Game	
Use Case ID	UC_AccessGame
Use Case Name	Access Game
Actor(s)	Visitor, Player
Trigger	User clicks Play or Launch Game button on the web platform
Preconditions	Web platform is online; game build is deployed
Postconditions	Game loads in browser or redirects to downloadable build
Main Flow	1. User clicks Play. 2. System authenticates user if required (UC_Authenticate). 3. System loads game build. 4. Game initializes and Main Menu is displayed. 5. Player stats are stored if applicable (UC_StoreStats).
Alternative Flow	If user is a Visitor without account, game may load in guest mode.
Exceptions	If game build fails to load, system displays error and retry option.

View Player Stats

UC_ViewStats — View Player Stats	
Use Case ID	UC_ViewStats
Use Case Name	View Player Stats
Actor(s)	Player
Trigger	Player clicks on Stats section while logged in
Preconditions	Player is authenticated; stats have been recorded from previous sessions
Postconditions	Player views personal gameplay statistics
Main Flow	1. Player clicks Stats. 2. System authenticates player (UC_Authenticate). 3. System retrieves player stats from database. 4. Stats page displays runs completed, enemies defeated, gold earned, upgrades purchased.
Alternative Flow	If no stats exist yet, system shows empty state with prompts to play.
Exceptions	If database is unreachable, system shows cached stats or error message.

View Leaderboard

UC_ViewLeaderboard — View Leaderboard	
Use Case ID	UC_ViewLeaderboard
Use Case Name	View Leaderboard
Actor(s)	Player
Trigger	Player clicks on Leaderboard section
Preconditions	Player is logged in; leaderboard data exists in the database
Postconditions	Player views ranked list of top performers
Main Flow	1. Player clicks Leaderboard. 2. System retrieves stored player stats (UC_StoreStats). 3. System ranks players by selected metric (e.g., hordes survived). 4. Leaderboard is displayed with player names and scores.
Alternative Flow	If player is not ranked yet, their row is shown at the bottom with current stats.
Exceptions	If database query fails, system displays last cached leaderboard with timestamp.

Manage Web Content

UC_ManageContent — Manage Web Content	
Use Case ID	UC_ManageContent
Use Case Name	Manage Web Content
Actor(s)	Admin
Trigger	Admin logs into the admin panel and selects Content Management
Preconditions	Admin is authenticated with admin-level credentials
Postconditions	Web content (overview, rules, credits) is updated and published
Main Flow	1. Admin logs in. 2. Admin navigates to Content Management section. 3. Admin edits text, images, or links. 4. Admin publishes changes. 5. System updates the live web pages.
Alternative Flow	Admin may save a draft before publishing.
Exceptions	If publish fails, system rolls back and notifies admin.

Manage Users

UC_ManageUsers — Manage Users	
Use Case ID	UC_ManageUsers
Use Case Name	Manage Users
Actor(s)	Admin
Trigger	Admin navigates to User Management section
Preconditions	Admin is authenticated (UC_Authenticate)
Postconditions	Admin has viewed, edited, or removed user accounts as needed
Main Flow	1. Admin opens User Management. 2. System authenticates admin (UC_Authenticate). 3. System displays user list with roles and activity. 4. Admin selects a user to view, edit role, or deactivate account. 5. Changes are saved to database.
Alternative Flow	Admin may filter users by role or activity status.
Exceptions	If user deletion fails due to active session, system notifies admin and cancels operation.

View Analytics

UC_ViewAnalytics — View Analytics	
Use Case ID	UC_ViewAnalytics
Use Case Name	View Analytics
Actor(s)	Admin
Trigger	Admin navigates to Analytics dashboard
Preconditions	Admin is authenticated; usage data has been collected
Postconditions	Admin views aggregated platform metrics
Main Flow	1. Admin opens Analytics section. 2. System retrieves aggregated data: active users, games played, most used cards, average horde survival. 3. Dashboard displays charts and summary stats.
Alternative Flow	Admin may filter by date range.
Exceptions	If analytics service is unavailable, system displays last known snapshot with timestamp.

Update Game Data

UC_UpdateGameData — Update Game Data	
Use Case ID	UC_UpdateGameData
Use Case Name	Update Game Data
Actor(s)	Admin
Trigger	Admin modifies game configuration data (card stats, enemy stats, level parameters)
Preconditions	Admin is authenticated (UC_Authenticate); game data schema is defined
Postconditions	Updated game data is stored and reflected in the live game on next load
Main Flow	1. Admin opens Game Data Management. 2. System authenticates admin (UC_Authenticate). 3. Admin selects data category (cards, enemies, levels). 4. Admin modifies values. 5. System validates changes against schema. 6. Changes are saved and published to game database.
Alternative Flow	Admin may preview changes before publishing.
Exceptions	If validation fails, system highlights invalid fields and prevents save.

Store Player Stats

UC_StoreStats — Store Player Stats	
Use Case ID	UC_StoreStats
Use Case Name	Store Player Stats
Actor(s)	Save System
Trigger	Player completes a horde, dies, or finishes a run
Preconditions	A gameplay session event has concluded
Postconditions	Player stats are persisted in the database for leaderboard and personal stats display
Main Flow	1. Gameplay event concludes (horde won, death, or victory). 2. System collects stat data: hordes survived, enemies defeated, gold earned, upgrades purchased. 3. System writes stats to

	player record in database. 4. Leaderboard rankings are recalculated.
Alternative Flow	If player is in guest mode, stats are stored locally only.
Exceptions	If database write fails, system queues stat update and retries on next connection.

Authenticate User

UC_Authenticate — Authenticate User	
Use Case ID	UC_Authenticate
Use Case Name	Authenticate User
Actor(s)	Player, Admin
Trigger	User attempts to access a protected feature (stats, leaderboard, admin panel)
Preconditions	User has an account; authentication service is online
Postconditions	User is granted or denied access based on credentials and role
Main Flow	1. User attempts to access protected feature. 2. System redirects to login screen if not authenticated. 3. User enters credentials. 4. System validates credentials against user database. 5. On success, session token is issued and user accesses feature.
Alternative Flow	If user has no account, system offers registration option.
Exceptions	If authentication service is down, system denies access and displays maintenance message.